

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electrical Engineering) SEMESTER - 6 Summer 2025 (Regular)

Course : Bachelor of Technology (Electrical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTEEC603 - CONTROL SYSTEM ENGINEERING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

1 Adding a zero to a transfer function generally

- | | |
|------------------------|--------------------------------|
| A) Increases overshoot | B) Reduces overshoot |
| C) Has no effect | D) Improves steady-state error |

Controllability in state-space means

- | | |
|--|--------------------------|
| A) Any state can be reached from any initial state with suitable input | B) States are observable |
| C) Poles are stable | D) System is stable |

3 For a type-1 system, the steady-state error for a unit ramp input is

- | | |
|-------------|-----------------------|
| A) Infinite | B) Zero |
| C) Finite | D) Depends on ζ |

4 A position control system has $K_p = 50$ and velocity constant $K_v = 10$ What is the type of the system?

- | | |
|-----------|-----------|
| A) Type 0 | B) Type 1 |
| C) Type 2 | D) Type 3 |

A second-order underdamped system has peak overshoot 16% . what is its damping ratio?

- | | | | |
|--------|--------|--------|--------|
| A) 0.5 | B) 0.6 | C) 0.7 | D) 0.8 |
|--------|--------|--------|--------|

6 A Bode plot shows a slope of -20 dB/dec after 100 rad/s. Which is true?

- | | |
|--------------------------|--------------------------|
| A) 1 pole at origin | B) 1 pole at 100 rad/s |
| C) 1 zero at 100 rad/s | D) 2 poles at origin |

7 The steady-state error for unit step input in a system with transfer function.

$$G(s) = \frac{S}{S+5}$$

- A) 1 B) 0 C) 5 D) 0.2

8 The integral part of a PID controller helps to

- A) reduce overshoot B) improve speed of response
C) eliminate steady-state error D) increase damping

9 The derivative part of a PID controller mainly improves

- A) steady-state error B) type of system
C) transient response (e.g., rise time) D) error constant

10 Simplify this block combination : Forward: $G_1=3$ feedback: $\square=0.5$, negative feedback

- A) $\frac{3}{1+1.5}$ C) $\frac{3}{1+0.5}$
B) $\frac{3}{1-0.5}$ D) $\frac{3}{1+3}$

11 A forward path gain is 4. There are two non-touching loops of gains $-0.3s$ and $-0.2s$. What is Δ ?

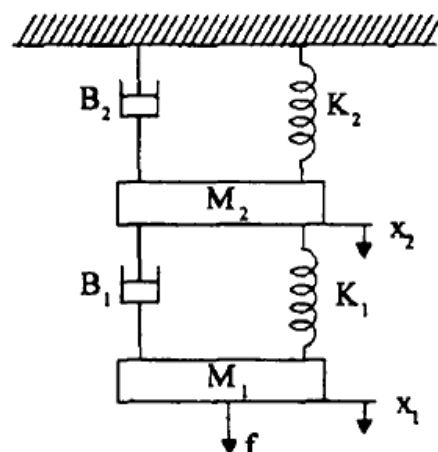
- A) $1 + 0.3s + 0.2s$ B) $1 - 0.3s - 0.2s$
C) $1 + 0.3s + 0.2s + 0.06s^2$ D) $1 - 0.3s - 0.2s + 0.06s^2$

12 Which operation is required to remove forward path summing?

- A) Take inverse B) Add blocks
C) Move summing point before the block D) Move summing point after the block

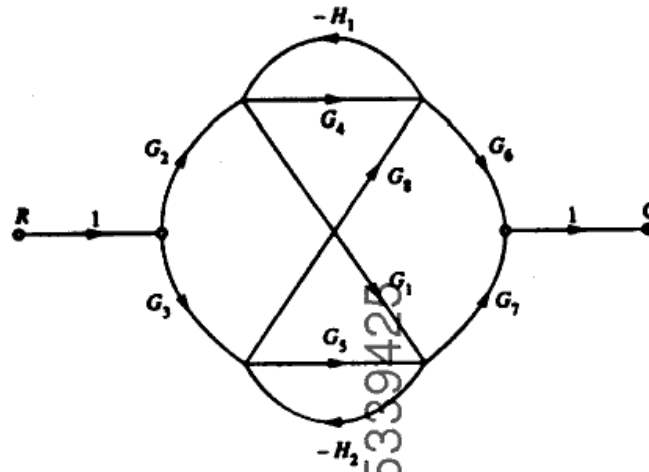
Q2. Solve the following.

A) Determine transfer function of given system.



6

- B) Determine transfer function of given system using signal flow graph method. 6



- Q3. Solve the following.

Find the steady state error for unit step, unit ramp and unit acceleration inputs for the following systems. 6

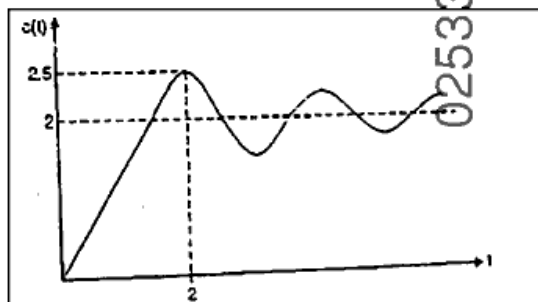
$$\frac{1000}{s^2(s+1)(s+20)}$$

- B) Explain Difference between open loop and closed loop control system 6

- Q4. Solve Any Two of the following.

A) Derive the expression for rise time and delay time of a unit-step response of a second-order system. 6

B) The step response of second order system is shown in figure. for an input of 2.5. Determine closed loop transfer function. assume unity feedback. 6



- C) Using Routh's stability criterion determine stability of given system. 6

$$S^6 + 3S^5 + 5S^4 + 9S^3 + 8S^2 + 6S + 4 = 0$$

- Q5. Solve Any Two of the following.

- A) sketch root locus of given OLTF and comment on value of K for stability. 6

$$G(S)H(S) = \frac{K}{S(S+3)(S+4)(S+6)}$$

- B) Determine gain margin and phase margin using bode plot and comment on stability. 6

$$G(s) = \frac{10}{s(1+0.5s)(1+0.01s)}$$

- Draw polar plot of given system and determine gain crossover and phase crossover frequency. $G(S)H(S) =$ 6

$$\frac{10}{(s+1)(s+3)}$$

Q6. Solve Any Two of the following.

- A) Explain PI and PD controller with its neat diagram. write down its transfer function for second order system. 6

- B) Obtain the transfer function of the system defined by 6

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \\ \dot{x}_3 \end{bmatrix} = \begin{bmatrix} -1 & 1 & 0 \\ 0 & -1 & 1 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} u$$

$$y = [1 \ 0 \ 0] \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

Comment on the complete state observability of the following systems. 6

$$\dot{X} = \begin{bmatrix} 1 & 1 & 2 \\ 0 & 2 & 0 \\ 0 & 0 & 2 \end{bmatrix} X \quad Y = [1 \ 0 \ 0] X$$

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